

International Quantum Online Exam 2026

Advanced Electronics, Photonics, Light and Quantum Mechanics Olympiad Exam

Instructions:

1. This exam consists of 90 multiple-choice, mostly calculation based questions.
 2. Each question has four options (A, B, C, D), only one of which is correct.
 3. Time: 1 hour.
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Section 1:

Electronics

1. A series RLC circuit has $R = 10\ \Omega$, $L = 100\ \text{mH}$, and $C = 10\ \mu\text{F}$. If the circuit is driven by a sinusoidal voltage source with a peak voltage of $10\ \text{V}$, what is the peak current at resonance?
A) $1\ \text{A}$
B) $10\ \text{A}$
C) $100\ \text{A}$
D) $0.1\ \text{A}$
2. An operational amplifier (Op-Amp) is configured as a non-inverting amplifier with $R_1 = 1\ \text{k}\Omega$ and $R_f = 9\ \text{k}\Omega$. If the input voltage is $0.5\ \text{V}$, what is the output voltage?
A) $4.5\ \text{V}$
B) $5\ \text{V}$
C) $0.5\ \text{V}$
D) $9\ \text{V}$
3. A silicon diode has a forward voltage drop of $0.7\ \text{V}$. If it is in series with a $100\ \Omega$ resistor and a $5\ \text{V}$ DC supply, what is the current flowing through the circuit?
A) $43\ \text{mA}$
B) $50\ \text{mA}$

C) 70 mA

D) 0 mA

4. A common-emitter BJT amplifier has a current gain (β) of 100. If the base current (I_B) is 100 μ A, what is the collector current (I_C)?
- A) 1 mA
B) 10 mA
C) 100 mA
D) 1 A
5. A 12 V Zener diode is used to regulate a 15 V input voltage. If the series resistor is 150 Ω , what is the minimum load current the Zener can handle while maintaining regulation, assuming a minimum Zener current of 5 mA?
- A) 15 mA
B) 20 mA
C) 25 mA
D) 30 mA
6. A 4-bit R-2R ladder Digital-to-Analog Converter (DAC) has a reference voltage of 5 V. If the digital input is 1011 (binary), what is the analog output voltage?
- A) 3.125 V
B) 3.4375 V
C) 4.0625 V
D) 4.375 V
7. A capacitor of 220 μ F is charged to 10 V. If it is discharged through a 1 k Ω resistor, what is the time constant of the circuit?
- A) 0.22 ms
B) 2.2 ms
C) 22 ms
D) 220 ms
8. An inductor of 50 mH is connected to an AC source with a frequency of 1 kHz. What is its inductive reactance (X_L)?

- A) $31.4\ \Omega$
- B) $314\ \Omega$
- C) $3.14\ \text{k}\Omega$
- D) $31.4\ \text{k}\Omega$

9. A full-wave bridge rectifier is supplied with a 24 V (RMS) sinusoidal AC voltage. Assuming ideal diodes, what is the approximate peak DC output voltage?

- A) 12 V
- B) 24 V
- C) 33.9 V
- D) 48 V

10. An Op-Amp integrator circuit has an input resistor of $10\ \text{k}\Omega$ and a feedback capacitor of $1\ \mu\text{F}$. If a constant input voltage of 1 V is applied, what is the rate of change of the output voltage (dV/dt)?

- A) $-0.1\ \text{V/s}$
- B) $-1\ \text{V/s}$
- C) $-10\ \text{V/s}$
- D) $-100\ \text{V/s}$

11. A JFET has an $I_{\text{DSS}} = 10\ \text{mA}$ and $V_{\text{P}} = -4\ \text{V}$. If $V_{\text{GS}} = -2\ \text{V}$, what is the drain current (I_{D}) using Shockley's equation?

- A) 2.5 mA
- B) 5 mA
- C) 7.5 mA
- D) 10 mA

12. A voltage divider consists of two resistors, $R_1 = 2\ \text{k}\Omega$ and $R_2 = 3\ \text{k}\Omega$, connected across a 10 V supply. What is the voltage across R_2 ?

- A) 4 V
- B) 5 V
- C) 6 V
- D) 7 V

13. A 555 timer is configured in a stable mode with $R_A = 10\text{ k}\Omega$, $R_B = 47\text{ k}\Omega$ and $C = 0.1\text{ }\mu\text{F}$. What is the approximate oscillation frequency?
- A) 100 Hz
 - B) 140 Hz
 - C) 180 Hz
 - D) 220 Hz
14. A power amplifier delivers 50 W to a $8\text{ }\Omega$ load. What is the RMS current flowing through the load?
- A) 2.5 A
 - B) 3.54
 - C) 5 A
 - D) 6.25 A
15. A logic gate circuit produces a HIGH output only when all its inputs are LOW. Which type of gate is it?
- A) AND
 - B) OR
 - C) NAND
 - D) NOR
16. A photodiode generates a current of $10\text{ }\mu\text{A}$ when exposed to light. If its responsivity is 0.5 A/W , what is the incident optical power?
- A) $5\text{ }\mu\text{W}$
 - B) $10\text{ }\mu\text{W}$
 - C) $20\text{ }\mu\text{W}$
 - D) $50\text{ }\mu\text{W}$
17. An analog-to-digital converter (ADC) has a resolution of 8 bits and a full-scale input voltage of 5 V. What is the step size (quantization level)?
- A) 19.53 mV

- B) 39.06 mV
- C) 78.125 mV
- D) 156.25 mV

18. A MOSFET has a threshold voltage (V_t) of 1 V and a transconductance parameter (k_n' (W/L)) of 1 mA/V^2 . If $V_{GS} = 3 \text{ V}$, what is the drain current (I_D) in the saturation region?

- A) 1 mA
- B) 2 mA
- C) 4 mA
- D) 8 mA

19. An Op-Amp differentiator circuit has an input capacitor of $0.1 \mu\text{F}$ and a feedback resistor of $100 \text{ k}\Omega$. If the input voltage changes at a rate of 0.5 V/ms , what is the output voltage?

- A) -0.5 V
- B) -5 V
- C) -50 V
- D) -500 V

20. A filter has a cutoff frequency of 1 kHz. If the input signal frequency is 100 Hz, and the filter is a first-order low-pass filter, what is the approximate gain (in dB) at 100 Hz?

- A) 0 dB
- B) -3 dB
- C) -10 dB
- D) -20 dB

21. A voltage source of 12 V is connected to a load resistor of 50Ω . If the internal resistance of the source is 10Ω , what is the power delivered to the load?

- A) 1.67 W
- B) 2 W
- C) 2.5 W
- D) 3.33 W

22. A common-drain (source follower) FET amplifier has a transconductance (g_m) of 5 mS and a load resistance (R_L) of 1 k Ω . What is the approximate voltage gain?
- A) 0.5
 - B) 0.83
 - C) 1
 - D) 5
23. A parallel RLC circuit has $R = 1$ k Ω , $L = 10$ mH, and $C = 100$ nF. What is the resonant frequency?
- A) 1.59 kHz
 - B) 15.9 kHz
 - C) 159 kHz
 - D) 1.59 MHz
24. A digital circuit uses a 3.3 V supply. If a logic HIGH is defined as > 2.0 V and a logic LOW as < 0.8 V, what is the noise margin for a HIGH input?
- A) 0.8 V
 - B) 1.3 V
 - C) 2.0 V
 - D) 2.5 V
25. A power supply has a no-load voltage of 12 V and a full-load voltage of 11.5 V when delivering 1 A. What is its voltage regulation percentage?
- A) 4.17%
 - B) 4.35%
 - C) 5%
 - D) 8.7%
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Section 2:

Photonics

1. A single-mode optical fiber has a core refractive index of 1.46 and a cladding refractive index of 1.455. What is the numerical aperture (NA) of the fiber?
A) 0.085
B) 0.121
C) 0.158
D) 0.205
2. A laser emits light with a wavelength of 632.8 nm. If the power output is 1 mW, how many photons are emitted per second?
A) $3.17 \times 10^{15} \text{ photons/s}$
B) $3.17 \times 10^{18} \text{ photons/s}$
C) $6.32 \times 10^{15} \text{ photons/s}$
D) $6.32 \times 10^{18} \text{ photons/s}$
3. An optical signal with a power of 10 mW is launched into an optical fiber. After propagating 10 km, the power is measured to be 1 mW. What is the attenuation of the fiber in dB/km?
A) 0.5 dB/km
B) 1 dB/km
C) 2 dB/km
D) 10 dB/km
4. A Fabry-Perot resonator has a length of 1 cm and mirrors with reflectivity $R = 0.99$. If the refractive index of the medium inside is 1.5, what is the Free Spectral Range (FSR) in frequency?
A) 5 GHz
B) 10 GHz
C) 15 GHz

D) 20 GHz

5. A photodiode has a responsivity of 0.8 A/W at a wavelength of 1550 nm. What is its quantum efficiency at this wavelength?
- A) 50%
B) 64%
C) 75%
D) 80%
6. A Mach-Zehnder interferometer is used as a phase modulator. If the refractive index of one arm changes by 10^{-5} over a length of 1 cm, and the wavelength of light is 1.55 μm , what is the phase shift introduced?
- A) 0.04 rad
B) 0.08 rad
C) 0.12 rad
D) 0.16 rad
7. A laser beam with a diameter of 1 mm has a divergence angle of 1 mrad. What will be the beam diameter after propagating 100 meters?
- A) 10 mm
B) 10.1 mm
C) 101 mm
D) 100 mm
8. A fiber Bragg grating (FBG) is designed to reflect light at 1550 nm. If the effective refractive index of the fiber core is 1.45, what is the grating period?
- A) 0.267 μm
B) 0.534 μm
C) 1.07 μm
D) 2.14 μm
9. An Erbium-Doped Fiber Amplifier (EDFA) provides a gain of 20 dB. If the input signal power is 10 μW , what is the output signal power?

- A) 100 μW
- B) 1 mW
- C) 10 mW
- D) 100 mW

10. What is the critical angle for light passing from glass ($n=1.5$) to air ($n=1$)?

- A) 30 degrees
- B) 41.8 degrees
- C) 45 degrees
- D) 60 degrees

11. A silicon photo detector has a band gap energy of 1.12 eV. What is the maximum wavelength of light it can detect?

- A) 0.8 μm
- B) 1.1 μm
- C) 1.5 μm
- D) 2.2 μm

12. A laser cavity has a length of 30 cm. If the refractive index of the gain medium is 1.0, what is the mode spacing (longitudinal mode separation) in frequency?

- A) 250 MHz
- B) 500 MHz
- C) 1 GHz
- D) 2 GHz

13. An optical fiber has a core diameter of 50 μm and a numerical aperture of 0.2. How many modes can it support at a wavelength of 850 nm?

- A) Approximately 100
- B) Approximately 200
- C) Approximately 300
- D) Approximately 400

14. A Pockels cell is used to rotate the polarization of light. If the half-wave voltage is 3 kV and an electric field of 1 kV/cm is applied across a crystal of 1 cm length, what is the induced phase retardation?
- A) $\pi/3$ rad
 - B) $2\pi/3$ rad
 - C) π rad
 - D) 2π rad
15. A 10 Gbps optical signal is transmitted over a fiber with a chromatic dispersion of 17 ps/(nm·km). What is the maximum transmission distance if the laser line width is 0.1 nm and the maximum allowed dispersion penalty is 200 ps?
- A) 100 km
 - B) 117 km
 - C) 150 km
 - D) 200 km
16. What is the power density of a 5 mW laser beam focused to a spot size of 10 μm diameter?
- A) $6.37 \times 10^7 \text{ W/m}^2$
 - B) $6.37 \times 10^9 \text{ W/m}^2$
 - C) $6.37 \times 10^{11} \text{ W/m}^2$
 - D) $6.37 \times 10^{13} \text{ W/m}^2$
17. A thin film with a refractive index of 1.38 is deposited on a glass substrate ($n=1.52$) to act as an anti-reflection coating for light at 550 nm. What is the minimum thickness of the film required?
- A) 50 nm
 - B) 100 nm
 - C) 150 nm
 - D) 200 nm
18. An optical amplifier has a noise figure of 5 dB. If the input signal-to-noise ratio (SNR) is 30 dB, what is the output SNR?
- A) 20 dB
 - B) 25 dB

- C) 30 dB
- D) 35 dB

19. A semiconductor laser has a gain coefficient of 50 cm^{-1} and internal losses of 10 cm^{-1} . If the cavity length is $200 \text{ }\mu\text{m}$ and the mirror reflectivities are 0.3 and 0.9, what is the threshold gain?
- A) 10 cm^{-1}
 - B) 20 cm^{-1}
 - C) 30 cm^{-1}
 - D) 40 cm^{-1}
20. What is the maximum bit rate for a 10 km optical fiber link with a total dispersion of 500 ps, assuming a non-return-to-zero (NRZ) format where the bit period should be at least twice the dispersion?
- A) 1 Gbps
 - B) 500 Mbps
 - C) 200 Mbps
 - D) 100 Mbps
21. A grating has 1000 lines/mm. If light of wavelength 600 nm is incident normally, what is the angle of diffraction for the first order maximum?
- A) 30 degrees
 - B) 36.87 degrees
 - C) 45 degrees
 - D) 60 degrees
22. A silicon waveguide has a cross-section of $220 \text{ nm} \times 500 \text{ nm}$. If the refractive index of silicon is 3.47 and the cladding is SiO_2 ($n=1.45$), what is the effective refractive index of the fundamental mode at 1550 nm (approximate value)?
- A) 1.5
 - B) 2.0
 - C) 2.5
 - D) 3.0

23. A quantum dot emits light at 800 nm. If the energy level spacing is 0.1 eV, what is the wavelength of light emitted if the energy level spacing doubles?
- A) 400 nm
 - B) 600 nm
 - C) 800 nm
 - D) 1600 nm
24. An optical filter has a bandwidth of 1 nm centered at 1550 nm. What is the bandwidth in frequency?
- A) 125 GHz
 - B) 166 GHz
 - C) 200 GHz
 - D) 250 GHz
25. A solar cell has an open-circuit voltage of 0.6 V and a short-circuit current of 5 A. If the fill factor is 0.75, what is the maximum power output?
- A) 1.5 W
 - B) 2.25 W
 - C) 3.0 W
 - D) 3.75 W
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Section 3:

Light

1. A light ray passes from air ($n=1$) into glass ($n=1.5$) at an angle of incidence of 30 degrees. What is the angle of refraction?
- A) 19.47 degrees
 - B) 20.00 degrees
 - C) 22.54 degrees
 - D) 48.59 degrees

2. A convex lens has a focal length of 20 cm. An object is placed 30 cm in front of the lens. What is the image distance and is the image real or virtual?
- A) 60 cm, Real
 - B) 60 cm, Virtual
 - C) -12 cm, Real
 - D) -12 cm, Virtual
3. In Young's double-slit experiment, the slits are separated by 0.1 mm and the screen is 1 meter away. If the wavelength of light is 500 nm, what is the distance between the central maximum and the first bright fringe?
- A) 2.5 mm
 - B) 5 mm
 - C) 10 mm
 - D) 15 mm
4. A polarizing filter reduces the intensity of unpolarized light by 50%. If two such filters are placed one after another with their transmission axes at 45 degrees to each other, what percentage of the original unpolarized light intensity will be transmitted?
- A) 12.5%
 - B) 25%
 - C) 37.5%
 - D) 50%
5. A prism has an apex angle of 60 degrees and a refractive index of 1.5. What is the minimum angle of deviation for a light ray passing through it?
- A) 20 degrees
 - B) 30 degrees
 - C) 37.18 degrees
 - D) 41.81 degrees
6. What is the energy of a single photon of green light with a wavelength of 550 nm? (Planck's constant $h = 6.626 \times 10^{-34}$ J·s, speed of light $c = 3 \times 10^8$ m/s)
- A) 3.61×10^{-19} J

- B) $3.61 \times 10^{-20} \text{ J}$
- C) $1.21 \times 10^{-27} \text{ J}$
- D) $1.21 \times 10^{-28} \text{ J}$

7. A concave mirror has a radius of curvature of 40 cm. An object is placed 30 cm in front of the mirror. What is the image distance and is the image real or virtual?
- A) 12 cm, Real
 - B) -12 cm, Virtual
 - C) 60 cm, Real
 - D) -60 cm, Virtual
8. A diffraction grating has 5000 lines/cm. If light of wavelength 600 nm is incident normally, what is the maximum order of diffraction visible?
- A) 1
 - B) 2
 - C) 3
 - D) 4
9. A light source emits 100 lumens of luminous flux. If it is placed at the center of a sphere with a radius of 1 meter, what is the illuminance on the inner surface of the sphere?
- A) 100 lux
 - B) 25 lux
 - C) 7.96 lux
 - D) 15.92 lux
10. A star is moving away from Earth at a speed of $0.01c$ (where c is the speed of light). If it emits light with a rest wavelength of 600 nm, what is the observed wavelength on Earth due to the Doppler effect?
- A) 594 nm
 - B) 600 nm
 - C) 606 nm
 - D) 612 nm

11. A thin film of oil ($n=1.4$) floats on water ($n=1.33$). If white light is incident normally, what is the minimum non-zero thickness of the oil film that will appear bright (constructive interference) for green light ($\lambda = 550 \text{ nm}$)?
- A) 98.2 nm
 - B) 196.4 nm
 - C) 275 nm
 - D) 392.8 nm
12. A microscope has an objective lens with a focal length of 5 mm and an eyepiece with a focal length of 25 mm. If the distance between the lenses is 150 mm, what is the total magnification?
- A) 100x
 - B) 200x
 - C) 250x
 - D) 300x
13. What is the critical angle for a diamond ($n=2.42$) in air ($n=1$)?
- A) 24.4 degrees
 - B) 30 degrees
 - C) 41.8 degrees
 - D) 45 degrees
14. A laser beam has a power of 10 mW and a wavelength of 632.8 nm. If the beam diameter is 1 mm, what is the intensity of the laser beam?
- A) $1.27 \times 10^4 \text{ W/m}^2$
 - B) $1.27 \times 10^5 \text{ W/m}^2$
 - C) $1.27 \times 10^6 \text{ W/m}^2$
 - D) $1.27 \times 10^7 \text{ W/m}^2$
15. Two coherent light sources are in phase and emit light with a wavelength of 600 nm. If they are separated by 0.2 mm, at what angle will the first dark fringe appear on a distant screen?
- A) 0.086 degrees
 - B) 0.172 degrees

- C) 0.258 degrees
- D) 0.344 degrees

16. A light wave has an electric field amplitude of 100 V/m. What is the intensity of the light wave?
(Permittivity of free space $\epsilon_0 = 8.85 \times 10^{-12}$ F/m, speed of light $c = 3 \times 10^8$ m/s)

- A) 13.27 W/m^2
- B) 26.55 W/m^2
- C) 53.1 W/m^2
- D) 106.2 W/m^2

17. A lens forms a real image that is twice the size of the object. If the object is placed 15 cm from the lens, what is the focal length of the lens?

- A) 5 cm
- B) 10 cm
- C) 15 cm
- D) 20 cm

18. A diffraction grating has 300 lines/mm. If it is illuminated by white light, what is the angular separation between the first-order maxima for blue light (450 nm) and red light (650 nm)?

- A) 3.45 degrees
- B) 4.12 degrees
- C) 5.89 degrees
- D) 6.78 degrees

19. What is the maximum possible efficiency of a solar cell if the bandgap of the semiconductor is 1.12 eV and the incident photon energy is 2 eV?

- A) 28%
- B) 56%
- C) 70%
- D) 100%

20. A Michelson interferometer is used to measure a small displacement. If 100 fringes pass a reference point when the mirror is moved, and the wavelength of light is 632.8 nm, what is the displacement?
- A) 31.64 μm
 - B) 63.28 μm
 - C) 126.56 μm
 - D) 253.12 μm
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Section 4:

Quantum Mechanics

1. An electron is accelerated through a potential difference of 100 V. What is its de Broglie wavelength? (Electron mass = 9.109×10^{-31} kg, elementary charge = 1.602×10^{-19} C, Planck's constant $h = 6.626 \times 10^{-34}$ J·s)
 - A) 0.122 nm
 - B) 0.245 nm
 - C) 1.22 nm
 - D) 2.45 nm

2. A particle is confined in a one-dimensional infinite potential well of width L. What is the energy of the ground state ($n=1$)?
 - A) $h^2 / (8mL^2)$
 - B) $h^2 / (2mL^2)$
 - C) $2h^2 / (8mL^2)$
 - D) $4h^2 / (8mL^2)$

3. The uncertainty in the position of an electron is 1 Å (10^{-10} m). What is the minimum uncertainty in its momentum? (Planck's constant $h = 6.626 \times 10^{-34}$ J·s)
 - A) $5.27 \times 10^{-25} \text{ kg} \cdot \text{m/s}$
 - B) $1.05 \times 10^{-24} \text{ kg} \cdot \text{m/s}$
 - C) $5.27 \times 10^{-24} \text{ kg} \cdot \text{m/s}$
 - D) $1.05 \times 10^{-23} \text{ kg} \cdot \text{m/s}$

4. A hydrogen atom is in the $n=3$ state. What are the possible values of the angular momentum quantum number (l)?
- A) 0, 1, 2
B) 1, 2, 3
C) 0, 1, 2, 3
D) 0, 1
5. What is the energy of a photon emitted when an electron in a hydrogen atom transitions from the $n=3$ state to the $n=1$ state? (Rydberg constant $R_H = 1.097 \times 10^7 \text{ m}^{-1}$, $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$, $c = 3 \times 10^8 \text{ m/s}$)
- A) 1.21 eV
B) 10.2 eV
C) 12.09 eV
D) 13.6 eV
6. If the wave function of a particle is given by $\Psi(x) = A \times e^{-ax^2}$, what is the probability of finding the particle between $x = -\infty$ and $x = +\infty$? (Assume A is the normalization constant)
- A) 0
B) 0.5
C) 1
D) Depends on 'a'
7. A quantum harmonic oscillator has a fundamental frequency of 10^{14} Hz . What is the energy of its first excited state ($n=1$)? ($h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$)
- A) $3.313 \times 10^{-20} \text{ J}$
B) $6.626 \times 10^{-20} \text{ J}$
C) $9.939 \times 10^{-20} \text{ J}$
D) $1.325 \times 10^{-19} \text{ J}$
8. What is the degeneracy of the $n=2$ energy level in a hydrogen atom, considering spin?
- A) 2
B) 4
C) 6

D) 8

9. An electron has a kinetic energy of 1 eV. What is its velocity? (Electron mass = $9.109 \times 10^{-31} \text{ kg}$, $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$)

A) $5.93 \times 10^5 \text{ m/s}$

B) $1.87 \times 10^6 \text{ m/s}$

C) $3.00 \times 10^8 \text{ m/s}$

D) $1.00 \times 10^6 \text{ m/s}$

10. A photon with energy 2 eV strikes a metal surface with a work function of 1.5 eV. What is the maximum kinetic energy of the emitted photoelectron?

A) 0.5 eV

B) 1.5 eV

C) 2.0 eV

D) 3.5 eV

11. What is the expectation value of position $\langle x \rangle$ for a particle in the ground state of an infinite potential well from $x=0$ to $x=L$?

A) 0

B) $L/4$

C) $L/2$

D) L

12. A particle has a wave function $\Psi(x) = \left(\frac{2}{L}\right)^{\frac{1}{2}} \times \sin\left(\frac{\pi x}{L}\right)$ for $0 < x < L$, and 0 otherwise. What is the probability of finding the particle in the region $0 < x < \frac{L}{2}$?

A) 0.25

B) 0.5

C) 0.75

D) 1

13. What is the orbital angular momentum of an electron in a p-orbital ($l=1$)?

A) 0

B) $\hbar$

C) $\sqrt{2} \hbar$

D) $\sqrt{3} \hbar$

14. The lifetime of an excited state is 10^{-8} seconds. What is the minimum uncertainty in the energy of this state? ($\hbar = 1.054 \times 10^{-34} \text{ J}\cdot\text{s}$)

A) $1.054 \times 10^{-42} \text{ J}$

B) $1.054 \times 10^{-26} \text{ J}$

C) $1.054 \times 10^{-20} \text{ J}$

D) $1.054 \times 10^{-16} \text{ J}$

15. A quantum system has two possible energy states, E_1 and E_2 . If the system is in a superposition state $|\Psi\rangle = \left(\frac{1}{\sqrt{2}}\right) |E_1\rangle + \left(\frac{1}{\sqrt{2}}\right) |E_2\rangle$, what is the probability of measuring energy E_1 ?

A) 0

B) 0.25

C) 0.5

D) 1

16. What is the wavelength of a photon that has an energy of 1.5 eV? ($h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$, $c = 3 \times 10^8 \text{ m/s}$, $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$)

A) 413 nm

B) 827 nm

C) 1240 nm

D) 1653 nm

17. A particle has a spin of $1/2$. What are the possible values of the spin magnetic quantum number (m_s)?

A) $-1/2, 1/2$

B) 0, 1

C) -1, 0, 1

D) $-1/2, 0, 1/2$

18. If a particle has a momentum of $10^{-24} \text{ kg}\cdot\text{m/s}$, what is its de Broglie wavelength? ($h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$)

A) $6.626 \times 10^{-10} \text{ m}$

B) $6.626 \times 10^{-9} \text{ m}$

C) $6.626 \times 10^{-8} \text{ m}$

D) $6.626 \times 10^{-7} \text{ m}$

19. What is the minimum energy required to excite a hydrogen atom from its ground state ($n=1$) to the first excited state ($n=2$)?

A) 1.51 eV

B) 3.40 eV

C) 10.2 eV

D) 13.6 eV

20. A quantum system is described by the Hamiltonian $H = \frac{p^2}{2m} + \left(\frac{1}{2}\right)kx^2$. What kind of system does this Hamiltonian represent?

A) Free particle

B) Particle in a box

C) Harmonic oscillator

D) Hydrogen atom

For questions, email: info.intl.neurosc@gmail.com

Good luck!
